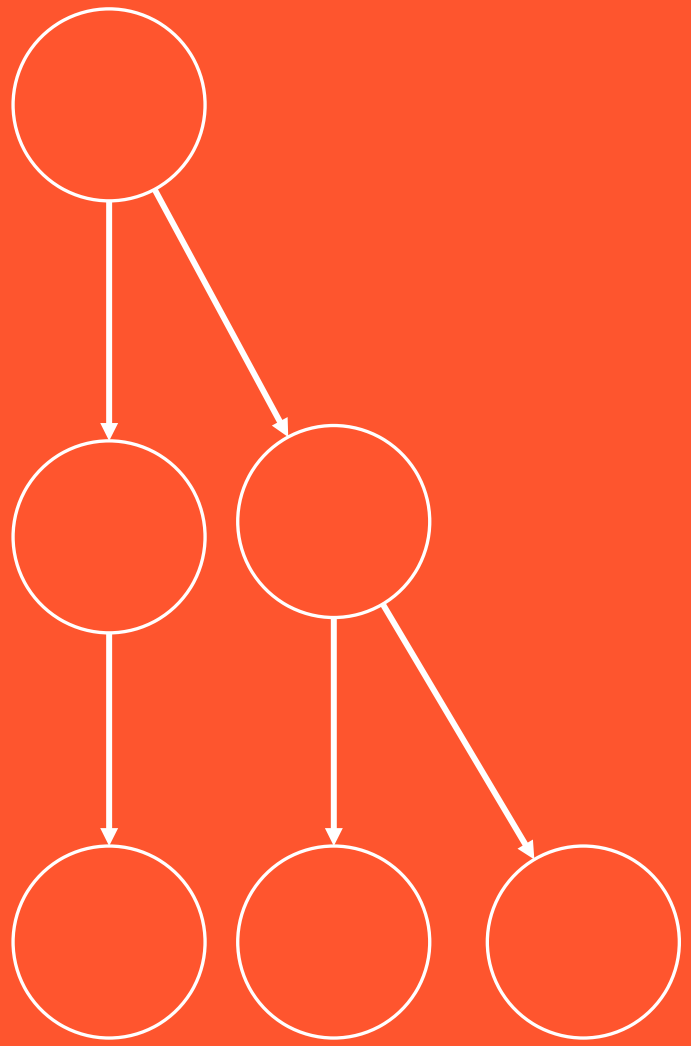


BCOG 100

Memory, Imagination, and Generative Models

Lecture 10.3 — Memory Systems in the Brain

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Remembered Differently

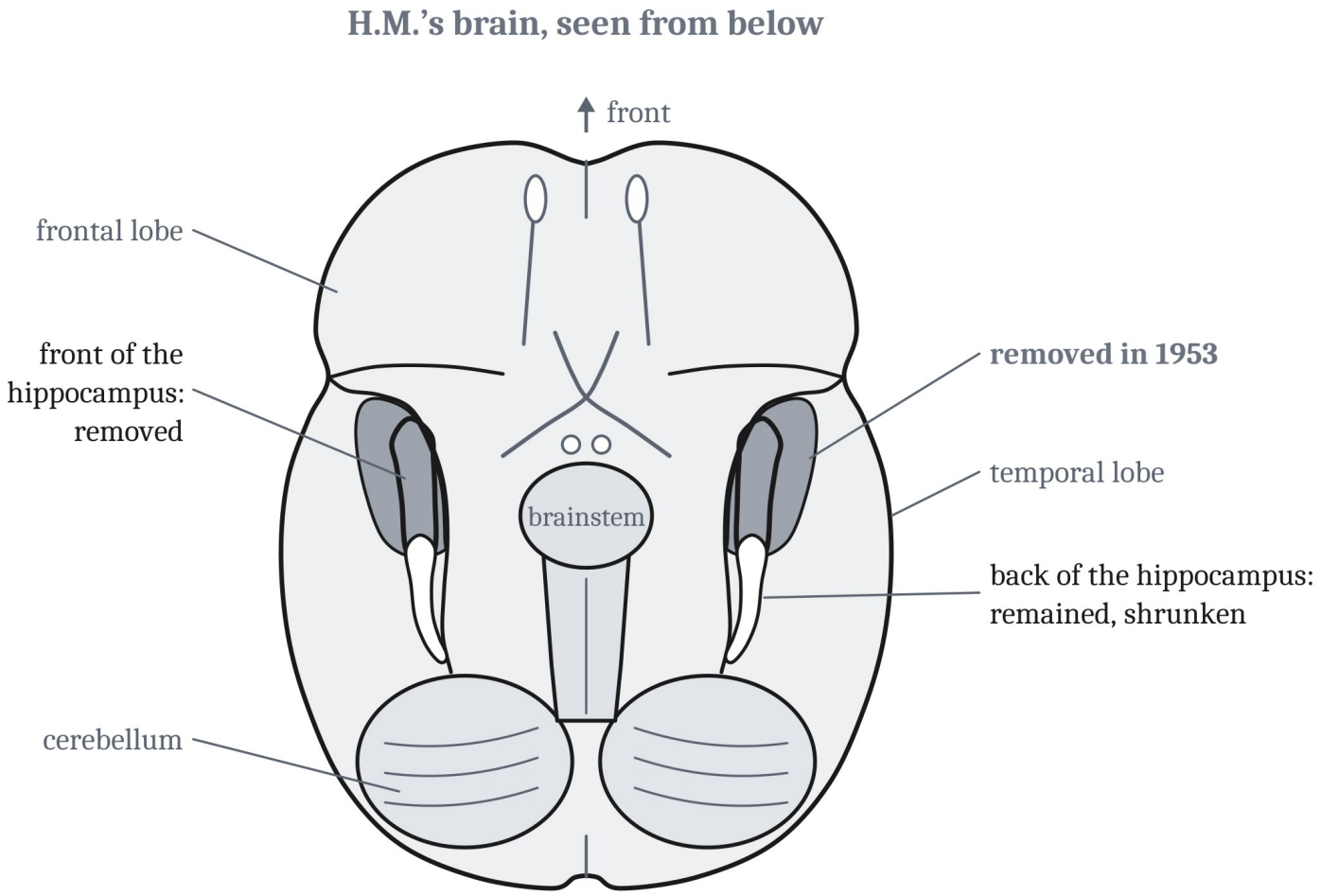
In your group, three minutes.

Think of an event you and someone else both remember, such as a family trip or a game, that the two of you remember differently.

What might each of you have filled in? Which of you is more confident, and does that tell you who is right?

H.M.

Amnesia: a severe loss of memory caused by brain damage or disease. A **dissociation** is the loss of one ability with another spared, after damage to one part of the brain.



schematic, after Corkin and colleagues (1997), as reproduced by Purves and colleagues (2018)

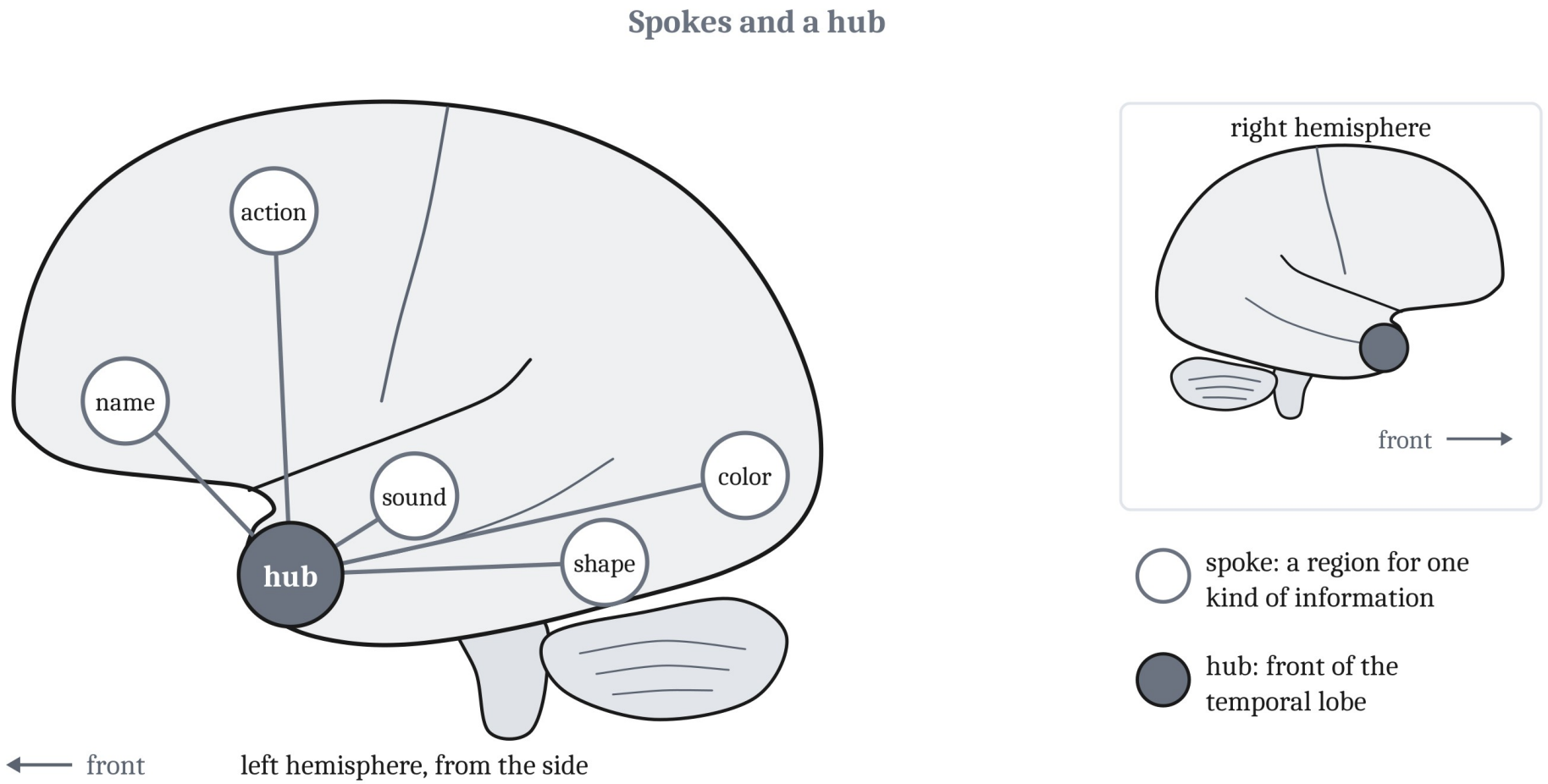
Losing the Past and the Future Together

Four amnesic patients, with different damage.

Patient	Damage	Lost	Spared
H.M.	Inner temporal lobes, both sides, including the front of the hippocampus	New facts and events; single past events	Intelligence, perception, language, learning skills (starting below normal)
R.B.	Cells in CA1, both sides, after blood flow was interrupted	New facts and events, profoundly	—
N.A.	Thalamus, from a fencing foil	New facts and events, less severely than H.M.	—
K.C.	Many regions, including the hippocampus, both sides	Every event of his life; new facts came slowly	Knowledge from before, including a machinist’s terms

The Semantic Hub

Semantic dementia destroys old knowledge of what things are and spares recent occasions: the opposite of hippocampal amnesia.

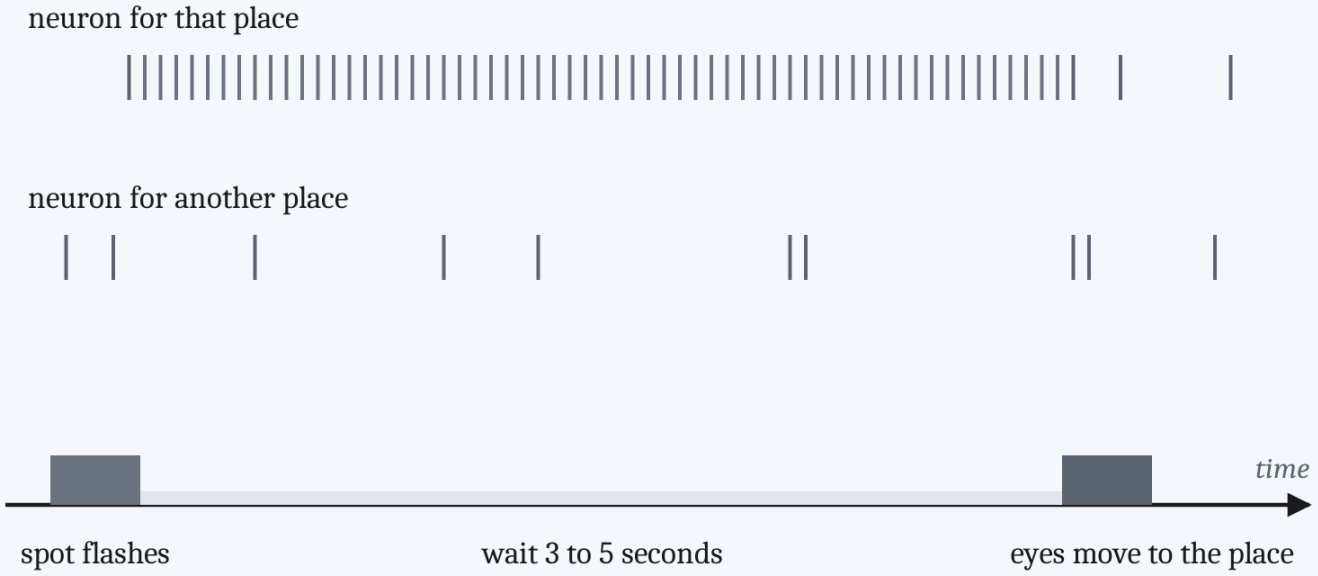


schematic, after Patterson and colleagues (2007) and Lambon Ralph and colleagues (2017); spoke positions approximate

Persistent Activity

Persistent activity: activity that continues through a delay after a stimulus has gone, and codes what must be remembered.

Remembering where a spot flashed



Prefrontal neurons fire through the wait, each most for one place or a few, always the same: its "memory field".

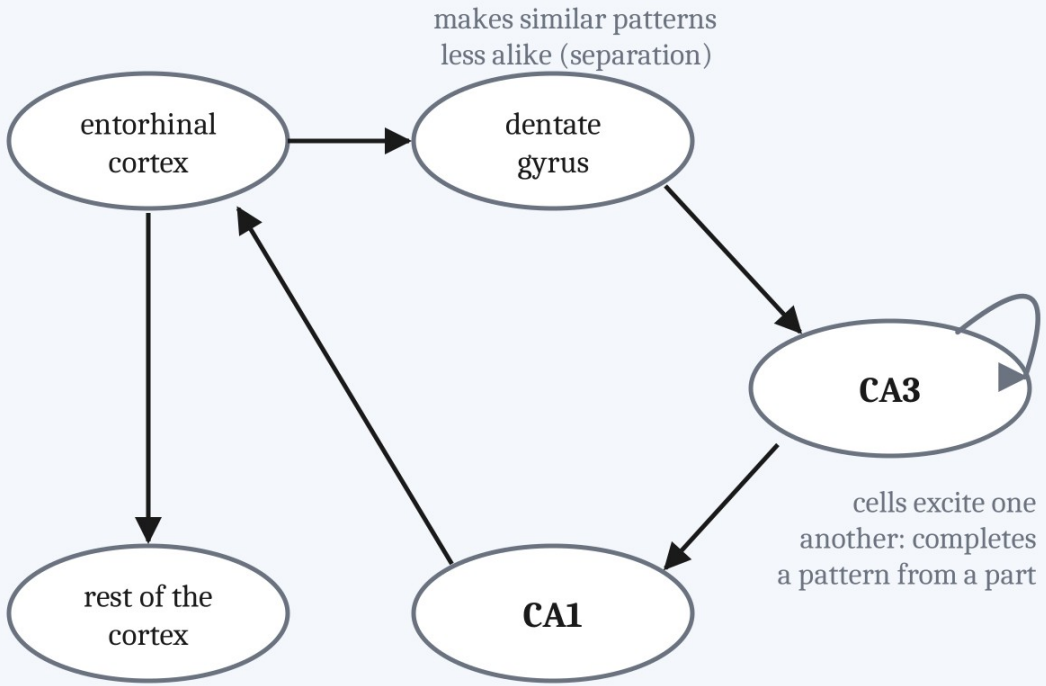
Other evidence complicates it

- When an attention task filled part of the wait, the delay activity vanished, yet the item was recalled correctly.
- On one proposal, short-lived changes in synapses then represent the item.
- The item can also be read from sensory, parietal, and temporal cortex.
- **For a few seconds, activity and short-lived changes in connections may both maintain what is remembered.**

Recall in the Circuit, and Engrams

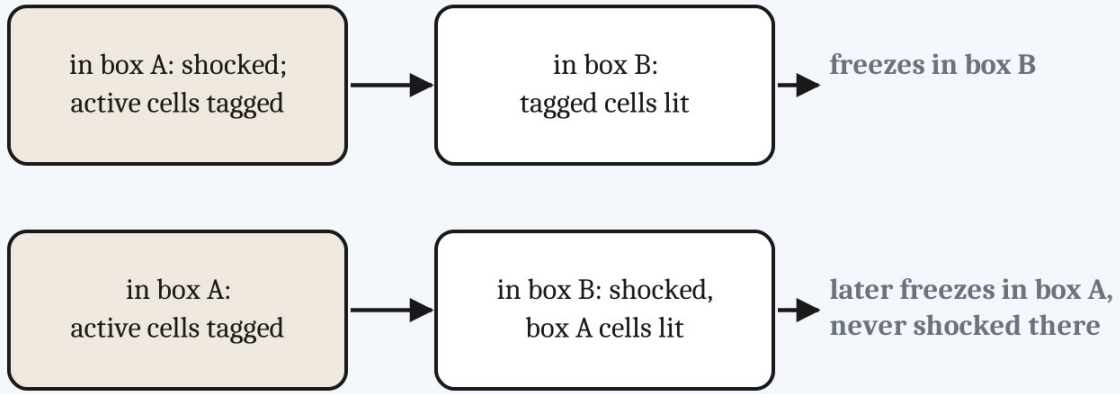
An **engram** is the lasting physical change that learning makes in the brain.

Recall in the loop, on one proposal



Engrams in mice

Cells active during fear learning are tagged,
so light can later reactivate them.



A small set of cells was enough to produce the
behavior of remembering.

Replay and Consolidation

A **sharp-wave ripple** is a brief burst of fast, synchronized hippocampal activity, about 40 to 120 milliseconds long.

Replay

On the track: place cells 1 to 6 fire in order, over several seconds



In slow-wave sleep, during a sharp-wave ripple (about 40 to 120 ms):

▪▪▪▪▪▪ **the same order, much faster**

Also awake and still, some in reverse. Disrupting ripples during sleep after maze training slowed learning.

Three accounts of consolidation

Standard

replay changes the cortex until a memory no longer needs the hippocampus; damage often impairs recent memories more than old

Multiple trace

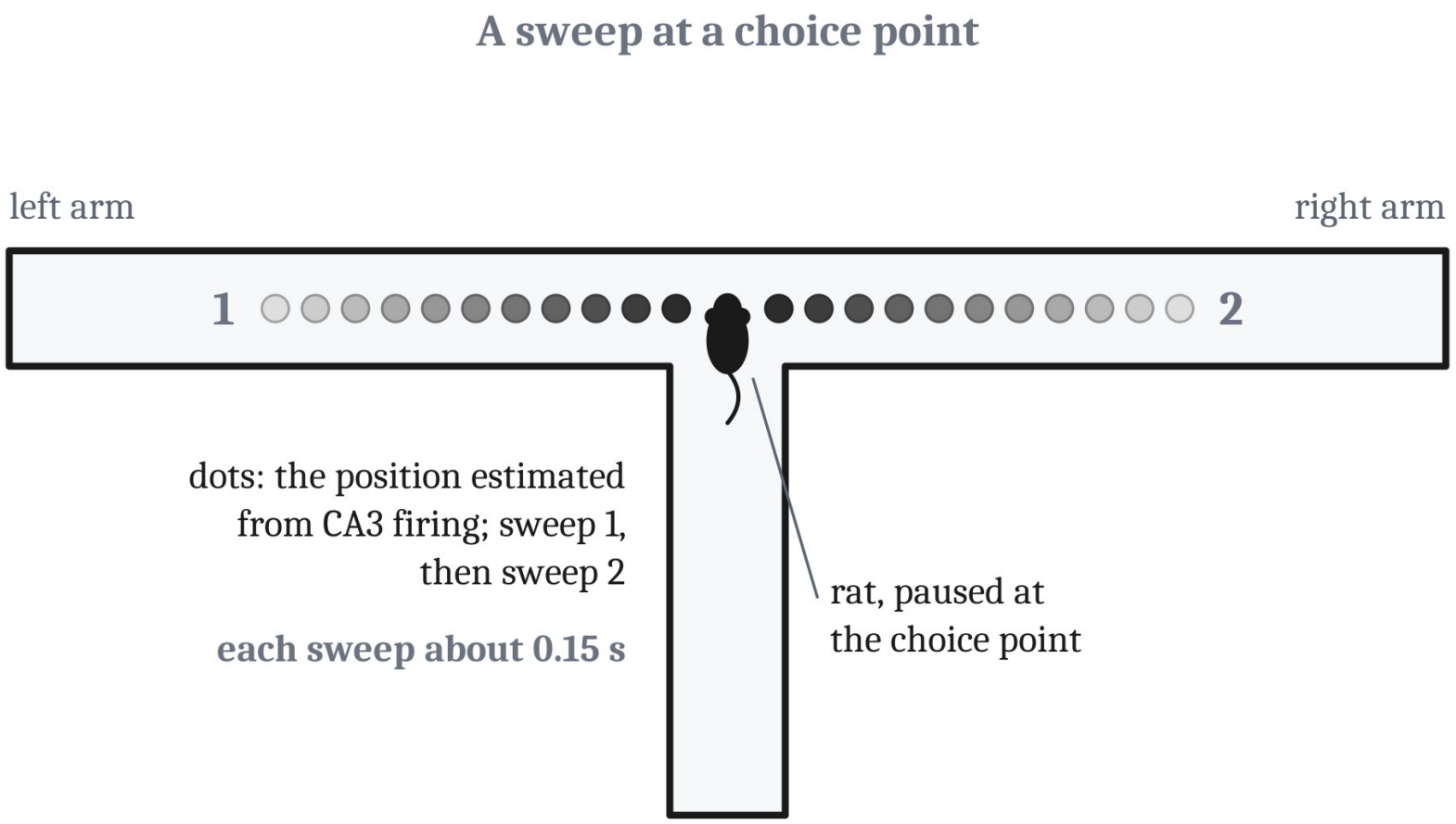
an occasion's details need the hippocampus as long as it is remembered; fits K.C., and H.M.'s old memories turned general

Trace transformation

a more general version comes to depend on the cortex; the detailed version still needs the hippocampus

Running Ahead at a Choice Point

When a rat paused at a fork, the place estimated from its CA3 cells sometimes shifted ahead, down one arm and then the other.



from above; schematic, after Johnson and Redish (2007), Fig. 7 and p. 12179

Remembering, Imagining, and the Default Network

The **default mode network**: regions more active when a person is not engaged by the outside world than during demanding tasks.

One core system

Remembering the past and imagining the future activate much the same regions:

- medial prefrontal cortex
- medial parietal cortex
- lateral temporal cortex
- medial temporal lobe, including the hippocampus

Imagining: frontal pole and hippocampus more active.
Remembering: visual regions more active.

The default mode network

Regions more active when a person is not engaged by the outside world than during demanding tasks.

Active in remembering, in language, in judging other people: not specific to remembering and imagining.

Imagining faces, houses, or chairs reactivates visual cortex that responds when they are seen; early visual cortex's part is still argued.

An event in parts

Two actors, two cafes, two grocery stores, recalled:

place
a cortical network

person
another cortical network

kind of situation
medial prefrontal cortex

the particular event
the hippocampus, closest

The cortex may represent reusable parts of events, recombined.

Infantile Amnesia

Infantile amnesia: adults recall almost nothing from their first years. Three proposed causes.

Too immature

An immature hippocampus or prefrontal cortex.

The human hippocampus grows rapidly until about age 2. Toddlers first find a place by its surroundings at about 21 months. In young rats, groups of place cells represent a path only in the third and fourth weeks.

Formed, not retrievable

Infant memories form but cannot later be retrieved.

In rodents, a fear learned in infancy and apparently forgotten returned after a reminder given up to four weeks later.

Not yet one's own past

Young children do not yet experience memories as their own past.

Fivush argues that children come to do so, by talking about the past with adults. Children whose mothers talked with them in detail about the past reported earlier first memories at age 12.

What May Be New in Mammals, and What the Hippocampus Is For

Memory for particular occasions, in some form, is probably not new in mammals. Three views of what the hippocampus is for remain open.

Probably not new in mammals

A hippocampal region has counterparts in fish, reptiles, and birds; scrub jays remember what, where, and when; cuttlefish, with no hippocampus, chose feeding places by the food and the time since they last ate. Claims about what mammals elaborated rest on correlations and proposals.

Map first

For

Hippocampal damage impairs place learning in goldfish, turtles, birds, and mammals; food-storing birds have a hippocampus over twice the expected size (a correlation)

Against

Nearly all comparative work tested places

Memory first

For

Silencing the hippocampus impaired picking odors by order, not new from old; the surprise fork, not the expected one: not perception, movement, or motivation

Against

In pigeons, songbirds, and turtles, lesions impair spatial learning "fairly selectively" in the tasks tested

Shared circuitry

For

Models store items, room maps, and sequences on one grid-cell scaffold; or treat finding a place as remembering relations

Against

Mostly untested predictions; one tested fit only modestly

Scenes and replay fit all three. In 2025, shared-circuitry advocates still called all three hypotheses.

Two Ways to Sort Memory Systems

The machines we described mostly store what they learn in one network, sometimes with one added store. The brain, on either scheme, has several systems.

Scheme	Kind of memory, or system	Depends on, or first appeared in
Standard	Declarative: facts and events	Medial temporal lobe, midline diencephalon
Standard	Nondeclarative: skills and habits	Striatum
Standard	Nondeclarative: priming, perceptual learning	Neocortex
Standard	Nondeclarative: simple conditioning	Amygdala, cerebellum
Accretion	Reinforcement	Early animals
Accretion	Navigation	Early vertebrates (in the hippocampus)
Accretion	Biased competition	Early mammals
Accretion	Manual foraging	Early primates
Accretion	Feature and goal	Anthropoid primates
Accretion	Social-subjective	The human line

To Discuss: Is Memory a Recording?

In your group, five minutes.

A friend says memory works like a video recording: an event is stored in one place, and played back when you remember it.

Using the evidence from this lecture, what would you tell them? Which piece of evidence would you start with?

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